ELECTROLYZER TECHNOLOGY

ISOTHERMAL ELECTROLYSIS.

>92% SYSTEM EFFICIENCY.

Pulsed waveform architecture eliminates waste heat. No platinum. No iridium. No membranes. Industrial-grade hydrogen production at unprecedented efficiency using commodity materials.

>92%	8×	<30°C	0
SYSTEM EFFICIENCY	LESS HEAT LOSS	OPERATING TEMP	RARE EARTH METALS

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STRATEGIC BRIEF

PUBLIC MASTER PLAN

Strategic Roadmap

> We move hydrogen—and by doing so, help usher in a future free of fossil fuels and carbon emissions. Traditional hydrogen is produced from fossil fuels (SMR) hundreds of miles away, liquefied for transport, and then driven by trucks to its final destination, all at enormous environmental and financial costs. We believe that's wildly inefficient and unsustainable. It's time we produced clean hydrogen locally, at ultra-high efficiency, using low-cost electricity and water.

OUR THREE-PHASE ROADMAP

Phase I: Build Close-Proximity Green Hydrogen Plants

Goal: Undercut costly, emission-heavy delivered hydrogen by producing it in regional facilities near industrial users.

- **Problem**: Hydrogen for Tulsa, OK, comes from Houston (fossil-based SMR), requiring liquefaction and **500+ mile trucking**—expensive, energy-intensive, and carbon-heavy.
- **Solution**:
  - Build smaller-scale green hydrogen plants *near demand centers* (eliminating transport/liquefaction costs).
  - Achieve cost parity with SMR where electricity costs ≤ **$0.13/kWh** (enabled by **>92% efficient** membrane-free electrolysis).
  - Avoid helium refrigeration complexities (reducing cost/carbon footprint).
- **Markets**: Start in Oklahoma; expand to Pacific Northwest, Great Plains, Desert Southwest (high H_2 demand, low electricity costs, abundant renewables).

Phase II: Replace Existing SMR with Electrolysis

Goal: Convert industrial hydrogen producers (oil refineries, petrochemical plants, steelmakers) from SMR to Tobe electrolyzers.

- **Why Industries Switch**:
  - Decarbonization pressure + carbon pricing make SMR increasingly expensive.
  - Tobe electrolyzers match SMR costs *without carbon emissions*.
  - Plug-and-play modular systems enable gradual SMR replacement (start small, scale up).
- **Impact**: Eliminate dependence on natural gas; qualify for clean energy incentives; serve "green hydrogen" premium markets.

Phase III: Integrate Hydrogen with Renewables for 24/7 Clean Energy

Goal: Use hydrogen as long-duration storage to enable fully renewable grids.

- **Mechanism**:
  - Excess solar/wind power → Tobe electrolyzers → hydrogen storage (hours to months).
  - Hydrogen → electricity via fuel cells during low-renewable periods.
- **Advantages over Batteries**:
  - Infinitely scalable, longer-duration, lower cost for grid balancing.
  - Critical for hospitals/data centers (replace diesel generators).
- **Outcome**: True 24/7 zero-carbon energy systems.

PUTTING IT ALL TOGETHER

Our plan is grounded in real economics and technology:

1. **BUILD CLOSE-PROXIMITY PLANTS** to out-compete delivered fossil hydrogen in targeted markets.
2. **REPLACE SMR** at large industrial sites, cutting CO_2 and aligning with global decarbonization trends.
3. **INTEGRATE WITH RENEWABLES** to create 24/7 clean energy systems and push the world closer to net zero.

> **THE KEY ENABLER**: Our high-efficiency, patent-pending membrane-free electrolyzers (**>92% SYSTEM EFFICIENCY**) allow cost-competitive green hydrogen at electricity prices ≤ **$0.13/KWH**—without carbon risk. We're building a future where green hydrogen is as commonplace—and invisible—as electricity is today. Join us in transforming industries, accelerating the clean energy transition, and preserving the planet.

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